

A trendline is a starting point

Follow-along worksheet

Your goal: explain a result and name a useful check.
TRY IT means type along. WATCH means follow the demo first.

Name

A pattern or forecast I want to check

Open follow-along.xlsx. Keep the existing sheet order. Windows: select the cell > Formulas > Insert Python. Ctrl+Enter runs the code.
Use Excel Value in the output menu to see numeric results in the grid.

1 Compare average fuel economy

TRY IT / mpg!K1. Type both lines and run.

```
mpg_df = xl("mpg[#All]", headers=True)
mpg_df.groupby("origin")["mpg"].mean()
```

Japan average minus Europe average (in MPG)

2 Put the gap in context

WATCH / mpg!K8. Welch's test compares the two group means.

```
from scipy import stats
japan = mpg_df.loc[mpg_df["origin"] == "japan", "mpg"].dropna()
europe = mpg_df.loc[mpg_df["origin"] == "europe", "mpg"].dropna()
test = stats.ttest_ind(japan, europe, equal_var=False)
pd.Series({"Gap in MPG": japan.mean() - europe.mean(),
          "p-value": test.pvalue})
```

Checks: gap about 2.56 MPG; p-value about 0.017. Interpretation continues on page 2.

A relationship you can explain

A small p-value is evidence against equal means under the test assumptions.

The p-value does not tell us the chance that the means are equal or whether the gap is useful. The test assumes independent observations. These historical practice data do not establish cause or a current car ranking.

3 Fit a line to the penguin data

TRY IT / penguins!!1. Insert Python and run this block.

```
penguins_df = xl("penguins[#All]", headers=True)
penguins_df = penguins_df.dropna(
    subset=["body_mass_g", "flipper_length_mm"])
plt.figure()
sns.regplot(data=penguins_df, x="body_mass_g",
            y="flipper_length_mm", ci=None, color="#CF3338")
```

Does the relationship go up or down?

Check: upward. We retain 342 of 344 rows with both measurements. Species and other differences may account for part of the pattern.

4 Give the slope a meaning

WATCH / penguins!!12. OLS fits the line numerically.

```
import statsmodels.api as sm
X = sm.add_constant(penguins_df[["body_mass_g"]])
model = sm.OLS(penguins_df["flipper_length_mm"], X).fit()
pd.Series({"mm per 100 g": 100 * model.params["body_mass_g"],
          "R-squared": model.rsquared})
```

Fitted extra flipper length for each extra 100 g of body mass (mm)

Check: about 1.53 mm. R-squared is about 0.759: the line accounts for about 76% of observed variation here. It is not future prediction accuracy.

A forecast needs a fair check

WATCH the forecasting steps. The code is here when you want to repeat them.

5 Read the monthly history

AirPassengers!D1. The passenger numbers are in thousands.

```
air = xl("passengers[#All]", headers=True)
air["date"] = pd.to_datetime(air["date"])
ts = air.set_index("date")["passengers"].sort_index()
ts = ts.asfreq("MS")
plt.figure()
ts.plot(color="#707070", ylabel="Passengers (000s)")
```

Trend = the longer-term direction. Seasonality = a repeating pattern.
This series has 144 months, January 1949 through December 1960.

6 Hide the last year, then predict it

AirPassengers!D8. Fit on 1949-1959. Predict 1960. Compare the model with simply repeating the previous year. MAE is average absolute error.

```
from statsmodels.tsa.holtwinters import ExponentialSmoothing
train, test = ts.iloc[:-12], ts.iloc[-12:]
fit = ExponentialSmoothing(train, trend="add",
    seasonal="add", seasonal_periods=12).fit()
pred = fit.forecast(12)
baseline = pd.Series(train.iloc[-12:].values, index=test.index)
pd.Series({"Repeat last year MAE": (test - baseline).abs().mean(),
    "Model MAE": (test - pred).abs().mean()})
```

Which method has the smaller average miss?

Result check

Repeat last year: about 47.8 thousand passengers MAE.
Holt-Winters model: about 13.4 thousand. Lower is better.
One successful test does not guarantee future accuracy.

The next 24 months

A forecast extends past patterns. Conditions can change.

7 Fit all the history and look ahead

WATCH / AirPassengers!D20. Run this after the page 3 steps.

```
final_fit = ExponentialSmoothing(ts, trend="add",
                                seasonal="add", seasonal_periods=12).fit()
future = final_fit.forecast(24)
plt.figure()
ts.plot(color="#707070", label="History")
future.plot(color="#CF3338", label="Forecast")
plt.ylabel("Passengers (000s)")
plt.legend()
```

The red line estimates 1961-1962. These are point estimates with no uncertainty interval shown. We used a simple additive seasonal model. Other models and additional test periods are useful next checks.

What past pattern is this forecast relying on?

Choose one result from today and explain it in plain English

What limitation or next check would you mention?

A useful explanation includes the result, its unit and a check.
Example: average miss of 13,400 passengers in the held-out year.

Your next step

Use one small example to answer a question from your own work.

A question I want to answer in my own workbook

A familiar result or simple baseline I will check against

When I will practice

One question I still have

If you get stuck

Use this worksheet's cell locations and run steps in order.

#PYTHON!: check names, quotes and whether earlier cells ran.

#SPILL!: clear space for the result. #BUSY!: wait for calculation.

An Image result is normal. Right-click it > Display Plot over Cells.

Keep building with the book

Python in Excel for Data Analytics

George Mount / Packt

Chapters 5-7: tests, regression and forecasting.

Work through examples and learn how to interpret and check the results.



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